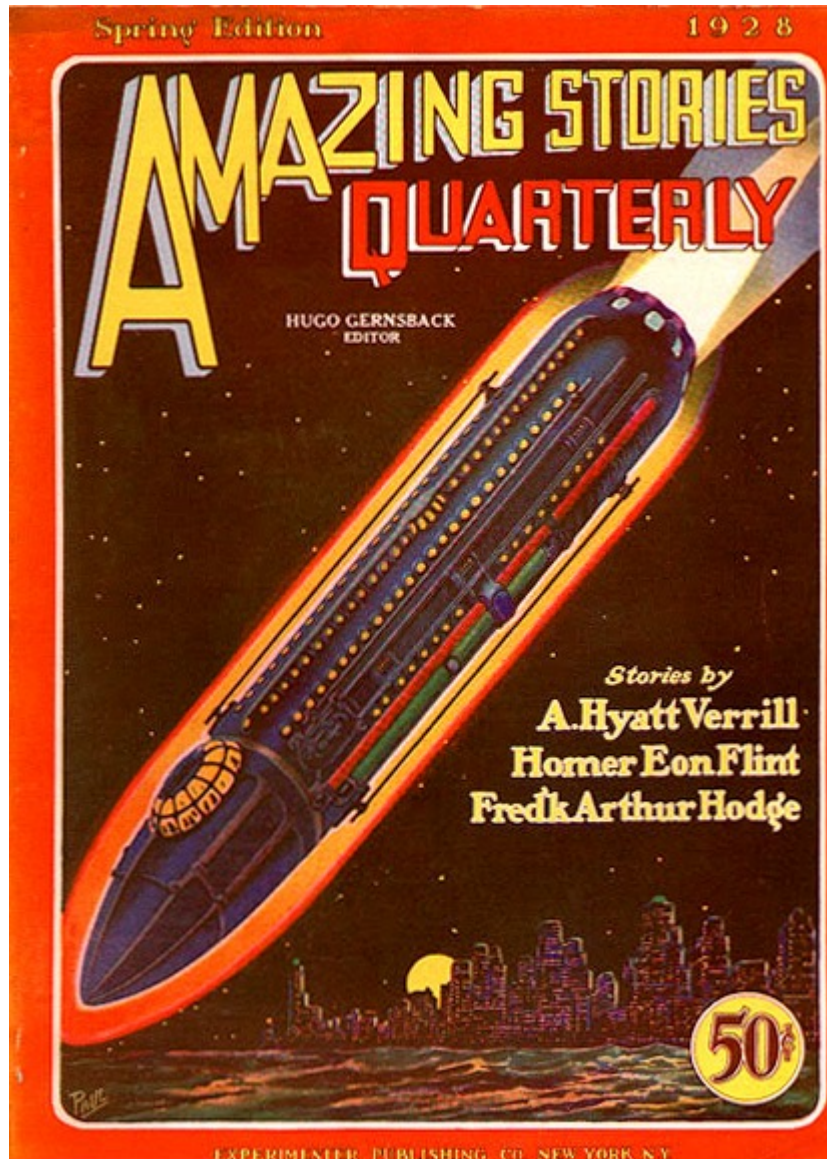




History of Science

Lecture 2: Creating computers

History of Science

Literature

Gerard Alberts en Bas van Vlijmen, *Computerpioniers*, Amsterdam: AUP (2017)

Gerard Alberts, De AERA. Gedroomde machines en de praktijk van het rekenwerk aan het Mathematisch Centrum te Amsterdam, in: *Studium* 1 (2008), 101-127

Gerard Alberts, Een halve eeuw computers in Nederland: een willekeurig getal, in: *Nieuwe Wiskrant* 22-1 (oktober 2002), 6-8

Slava Gerovitch, 'Mathematical Machines' of the Cold War: Soviet computing, American cybernetics and ideological disputes in the early 1950s, in: *Social Studies of Science* 31 (2001), 253-287

Petri Paju and Helena Durnová, Computing close to the Iron Curtain: Inter/national computing practices in Czechoslovakia and Finland, 1945-1970, in: *Comparative Technology Transfer and Society* 7 (2009), 303-322

Helena Durnová, Sovietization of Czechoslovakian computing: the rise and fall of the SAPO project, in: *IEEE Annals of the History of Computing* 32 (2010), 21-31

Matteo Pasquinelli, *The eye of the master. A social history of artificial intelligence*, London / New York (2023)

Andrew Pickering, *The cybernetic brain. Sketches of another future*, Chicago / London: UCP (2010)

Stephanie Dick, Artificial Intelligence, in: *Harvard Data Science Review* 1.1 (2019), <https://doi.org/10.1162/99608f92.92fe150c>

Stephanie Dick, The Marxist in the machine, in: *Osiris* 38 (2023), 61-81

History of Science

Overview:

- Cold War Science
- (Dutch) Dinosaurs or: what was the first computer?
- AI and cultural dispositions

History of Science

Today we will focus on the ideals that made people bother. Why after WW II people started putting faith in machines that evidently did not work.

Meanwhile we will discuss how the question “which one was the first computer” became relevant.

Z1 (1938)



History of Science

Cold War Science

But let us start by clearing some prejudice.

Marian Adam Rejewski (1905-1980), Jerzy Witold Różycki (1909-1942), Henryk Zygalski (1908-1978)



History of Science

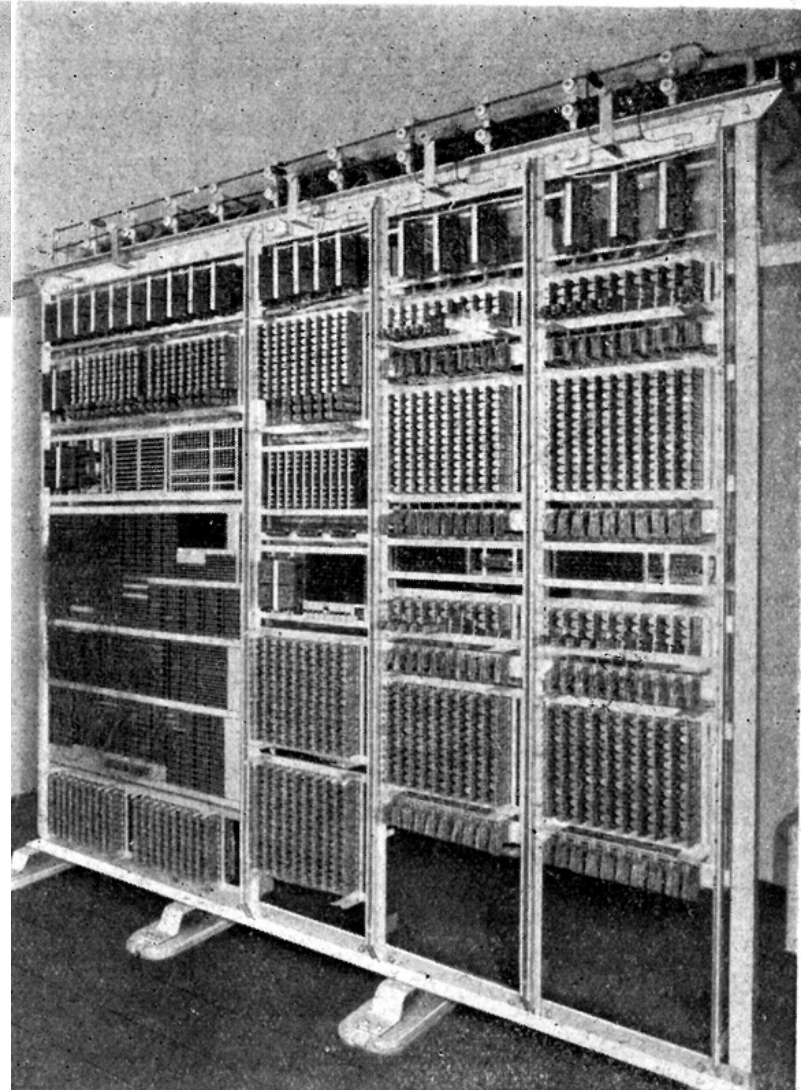
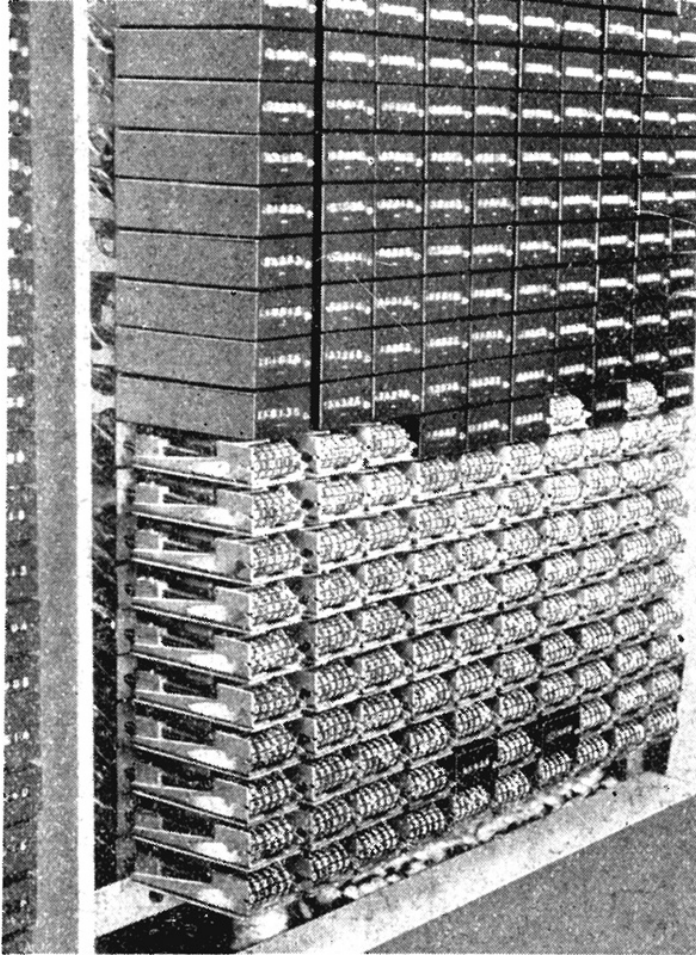
Cold War Science

More prejudice.

Campbell-Kelly, page 89: “Britain was the only country other than the United States not so devastated by the war that it could establish a serious computer research program.”



History of Science



Yamashita type statistical machine, 1944-1948

History of Science

Intermezzo

The simple stories:

“Computers were invented because they made our lives easier.”

“Computers made things more efficient”

“Computers were invented by computer scientists”

“Computers were trusted from the beginning”

“Computers soon became a necessity”

Are based on prejudice and ill informed.

History of Science

Cold War Science

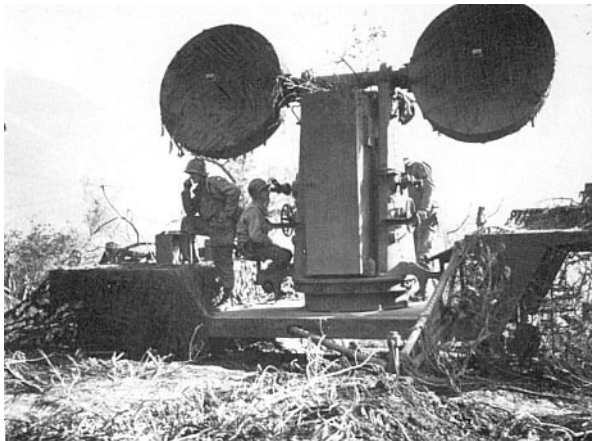
Manhattan Project

Aeronautical calculations

Code breaking (Bletchly Park)

Radar

Supplying troops



History of Science

Cold War Science

After WW II need for scientific calculations exploded

NASA

Atomic power



History of Science

Cold War Science

In (continental) Europe of the late 1940s as well:
rebuilding
the nation
(Marshall plan)

Mix of fatalism
and optimism

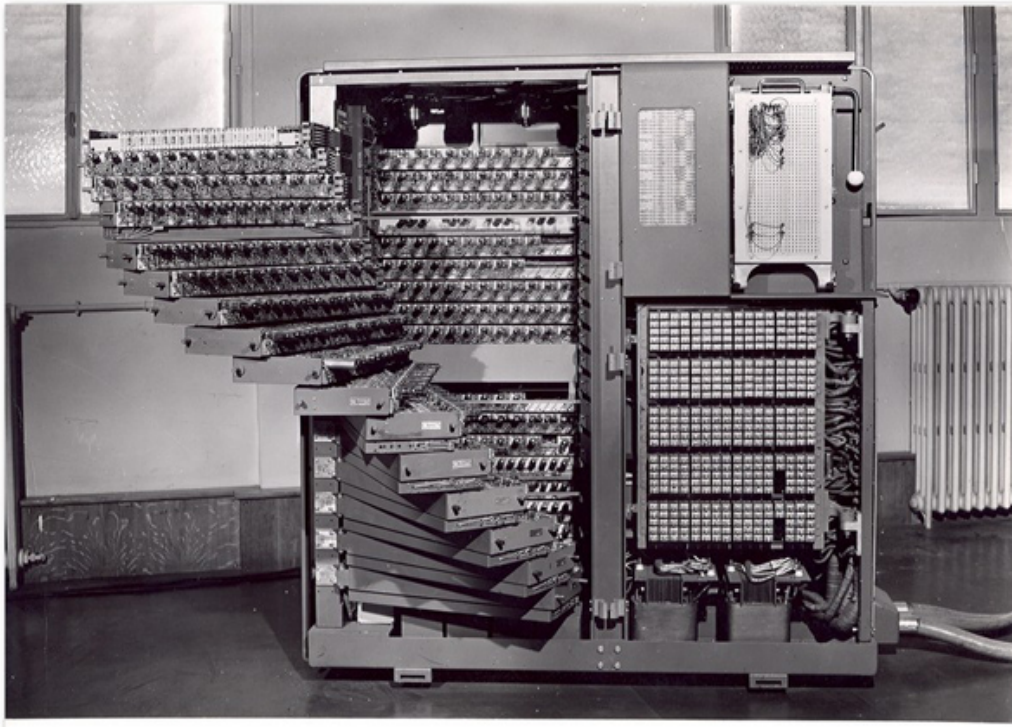


History of Science

Cold War Science

Jean Jacques Servan-Schreiber (1924-2006)

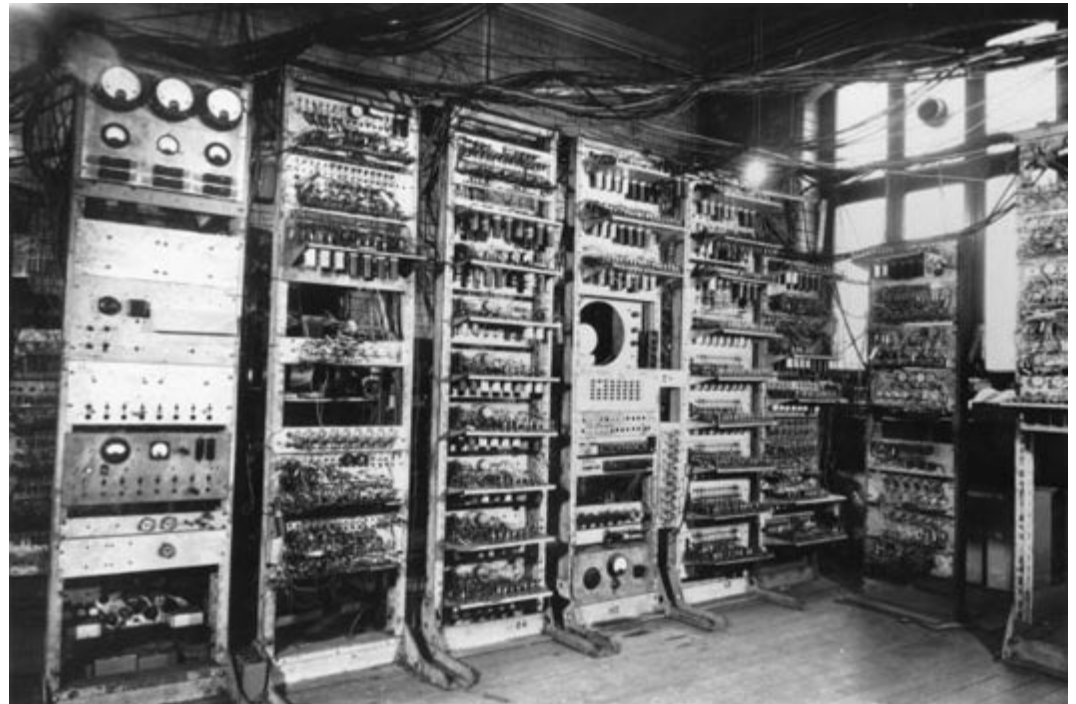
Le défis Américain (1968)



History of Science

Cold War Science

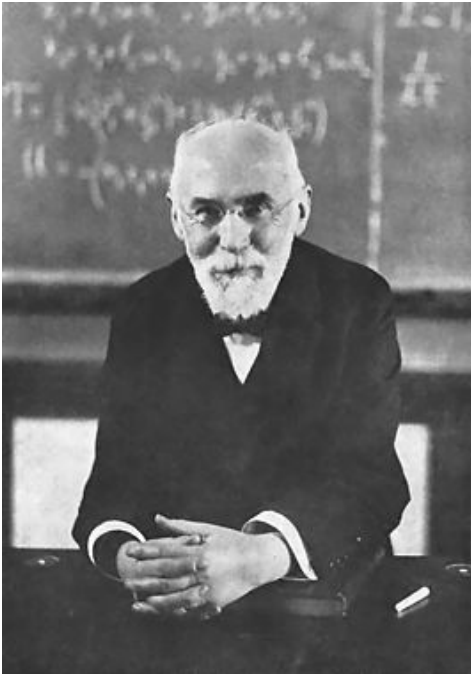
Manchester, december 1948: the “Manchester Baby”. War trodden Europe was quite ready to move into the computer age, as was Japan, as was Germany! The devastation could even be profitable: relays and radio tubes were relatively cheap.



History of Science

Dutch dinosaurs:

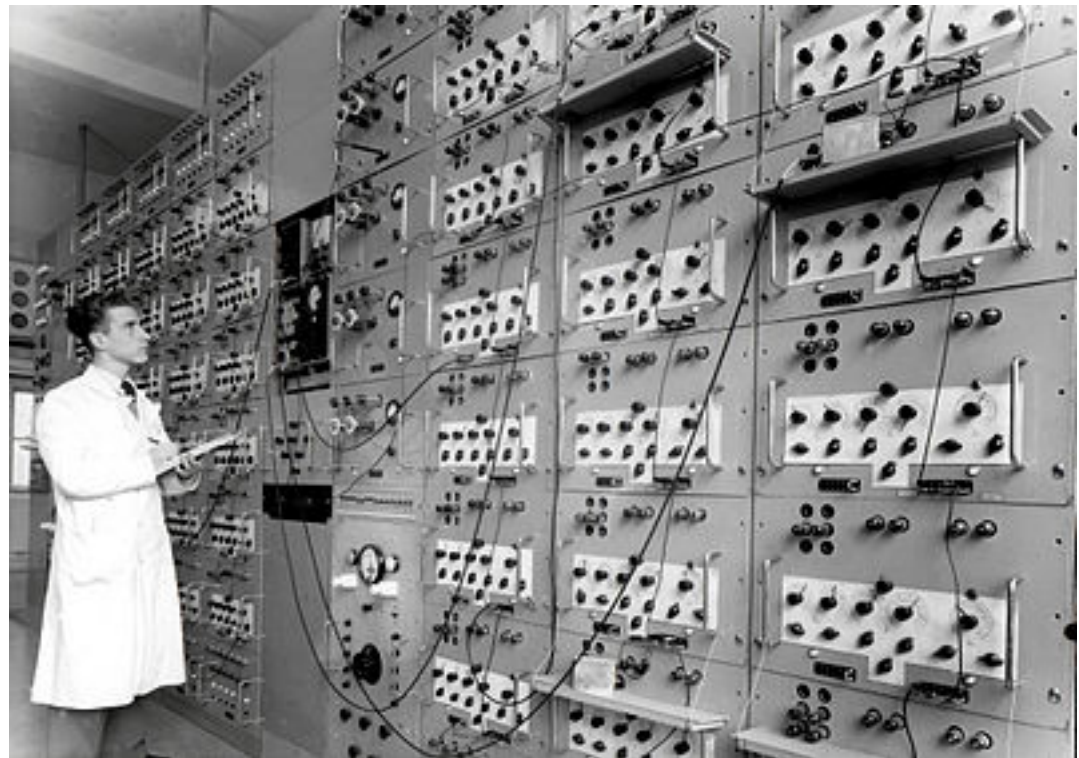
In the land reclamation and waterworks, huge calculation efforts were undertaken in the 1910s and 1920s by H.A. Lorentz and C. Lely.



History of Science

Dinosaurs:

In the tradition of process control: engineer Johan van Veen (1893-1953), Rijkswaterstaat



History of Science

Dinosaurs:

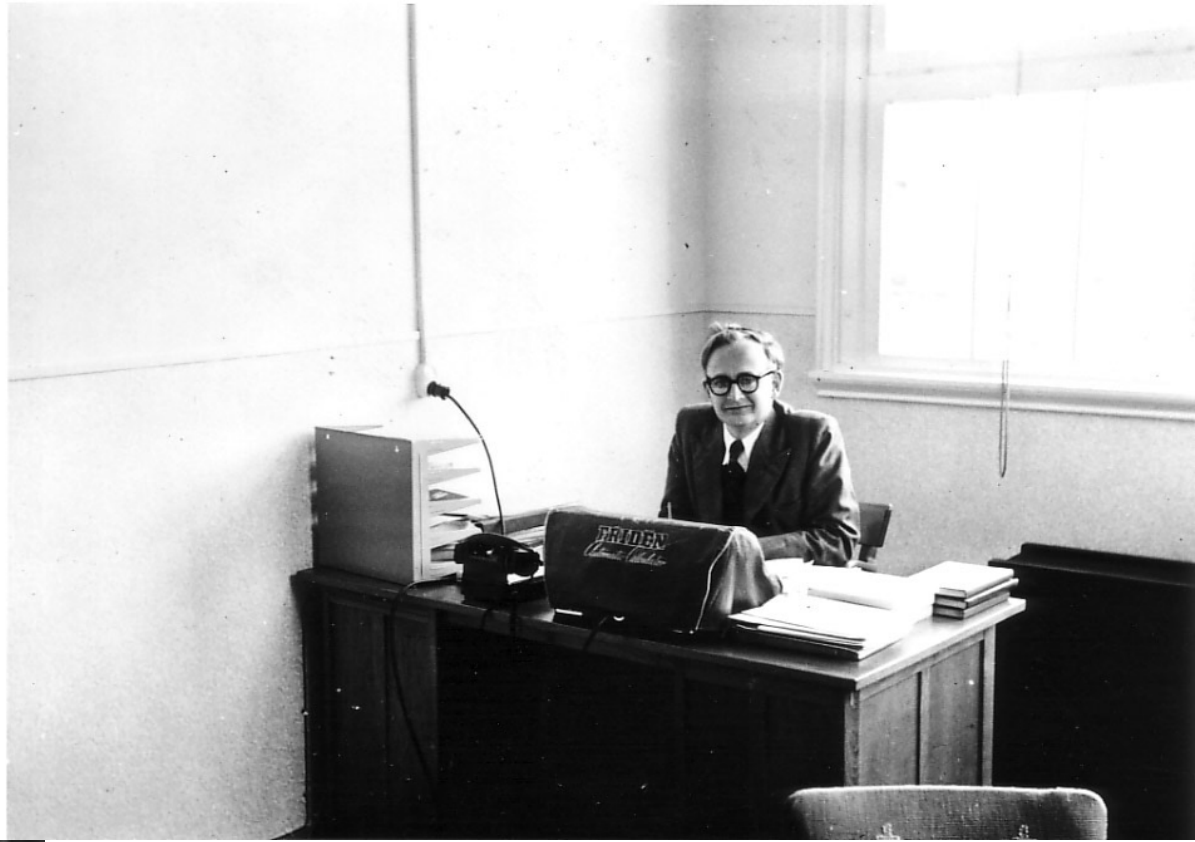
In the tradition of process control until 1972: Deltar



History of Science

Dinosaurs: Amsterdam

Mathematisch Centrum (1946) - Moore school



History of Science

Dinosaurs: Amsterdam

b	NATIONAAL LUCHTVAARTLABORATORIUM		
	<u>Verslag van de engelsche reis van Dr.Ir.A.van Wijngaarden</u> <u>van 9-1-1946 tot 6-3-1946.</u>		

Verslag van de engelsche reis
van Dr. Ir. A.van Wijngaarden
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b	NATIONAAL LUCHTVAARTLABORATORIUM		
	<u>Verslag van de engelsche reis van Dr.Ir.A.van Wijngaarden</u> <u>van 9-1-1946 tot 6-3-1946.</u>		

In opdracht van Prof.Dr.Ir.G.B.Biezeno, daartoe in de gelegenheid gesteld door het Hoogeschoolfonds, werd een studiereis ondernomen naar Engeland. Hieronder volgt een kort verslag van de resultaten voornamelijk op het gebied der Toegepaste Mechanica.

Voornaamste bezochte laboratoria, instellingen en personen.

National Physical Laboratory (NPL), Teddington. Bezoekt in diverse bezoeken:

- Aero Division
- Ship Division (Kent, Wigley)
- Engineering Division (Cox)

History of Science

Dinosaurs: Amsterdam

Aad van Wijngaarden (1916-1987), from 1947 onwards director of the “computing department” of the Mathematisch Centrum



History of Science

Dinosaurs: Amsterdam

June 21, 1952: ARRA



History of Science

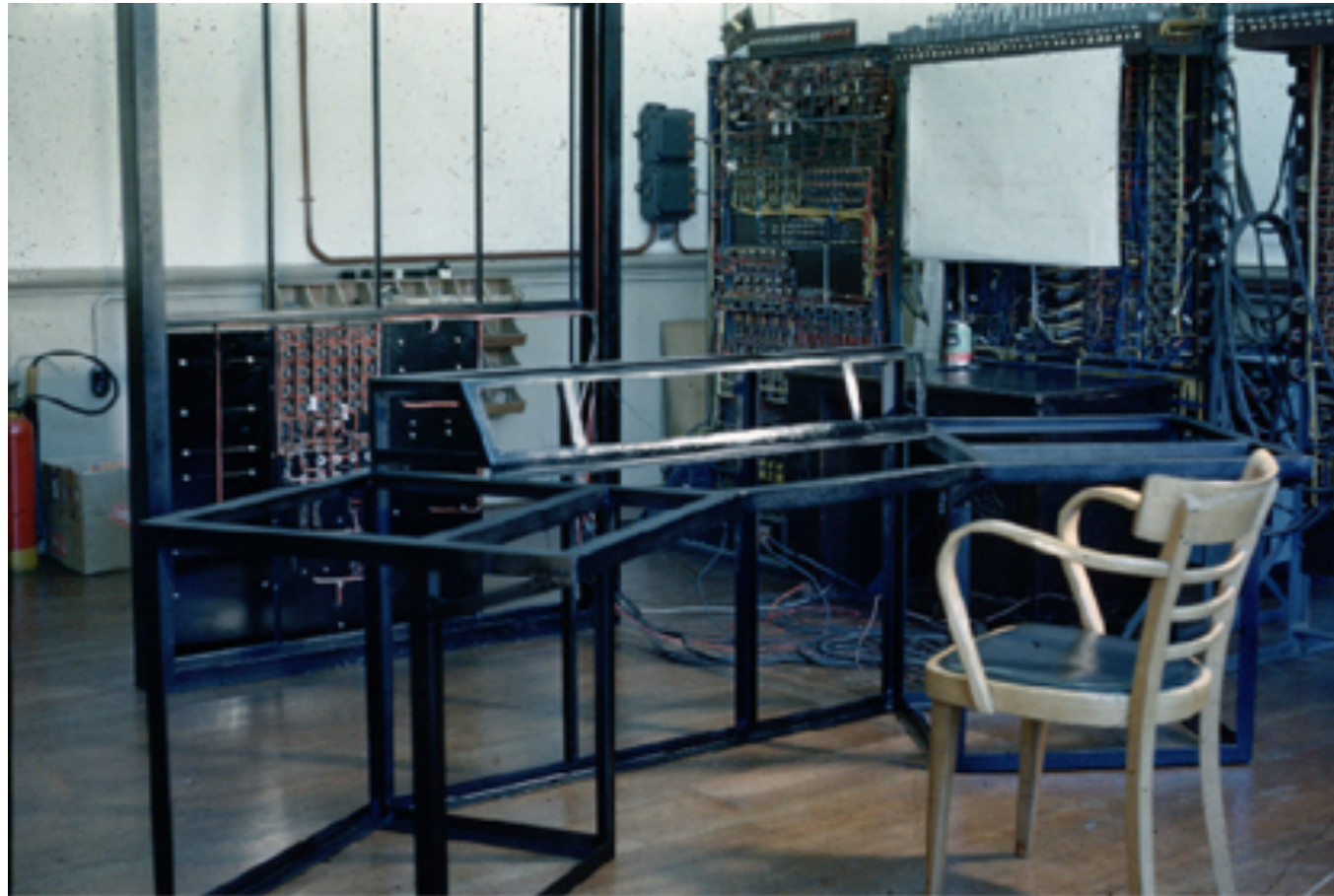
Dinosaurs: Amsterdam



History of Science

Dinosaurs: Amsterdam

“Bluffing
their way into
the computer
age” - Gerard
Alberts



History of Science

Dinosaurs: Amsterdam



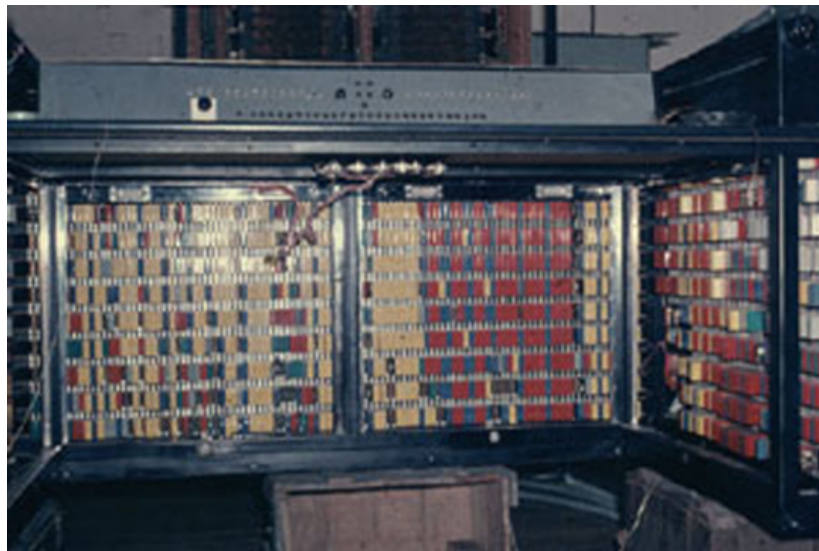
ARRA II

March 1954

History of Science

Dinosaurs: Amsterdam

Electrologica, spin off 1956
X1, subsequently bought by
Mathematisch Centrum



History of Science

Intermezzo: European dwarves

Electrologica (1956) - Dutch dwarf

Zuse KG (1949) - German dwarf

Regnecentralen (1955) - Danish dwarf

Bull (1931) - French dwarf

LEO (1949) - British dwarf



History of Science

Dinosaurs

Mathematisch

Centrum

jaar	R	O	MC	ARRA	ARRA II	ARMAC	X1
1946							
1947							
1948							
1949	64	9					
1950	114	52	27	2			
1951	166	59	21	1			
1952	206	48	17	1			
1953	249	52	13				
1954	297	59	8		8		
1955	340	53	13		20		
1956	394	60	9		5	13	
1957	467	73	9			38	
1958	514	57	5			28	
1959	557	55	6			17	2
1960	605	69	7			1	42
1961	725	122	11				122
1962	852	179*					179

History of Science

Dinosaurs: Delft

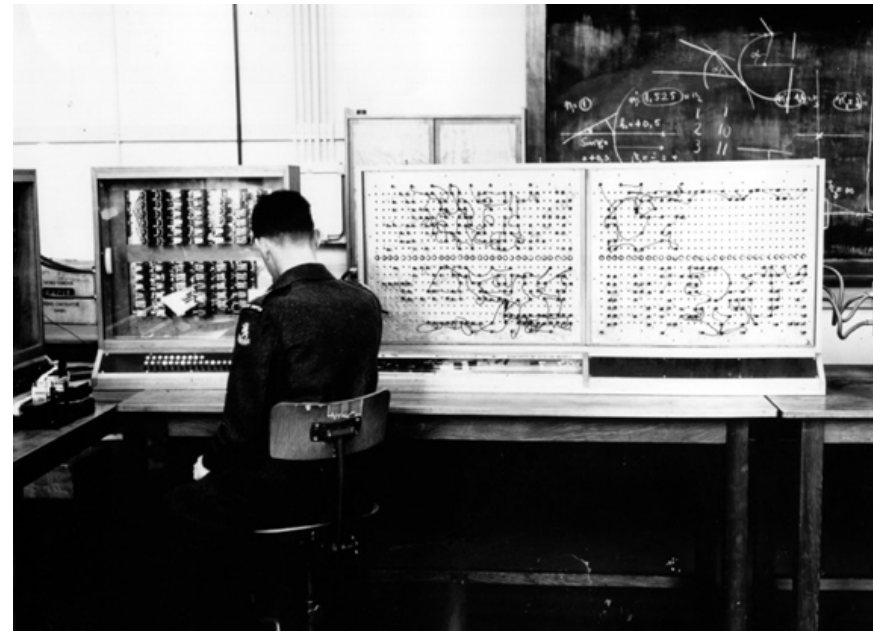
Willem van der Poel (*1926), school of prof.dr. N.G. de Bruijn (optics and fiber technology)

1950 ARCO (Automatic Relais Calculator for Optical problems)

later called: *Testudo*

Relays

Programs hard-wired,
but interchangeable



History of Science

Dinosaurs: Delft

Facilitating telephone traffic (PTT):

ZERO (Zeer Eenvoudig Reken Orgaan), 1951

PTERA (PTT Electronische Reken Automaat), 1953



History of Science

Dinosaurs: Delft

Beginning of 1955 Van der Poel designed ZEBRA
(Zeer Eenvoudig Binair Reken Apparaat)

In production 1957: Standard Telephone and Cable
Company (STANTEC)



History of Science

Dinosaurs: Eindhoven

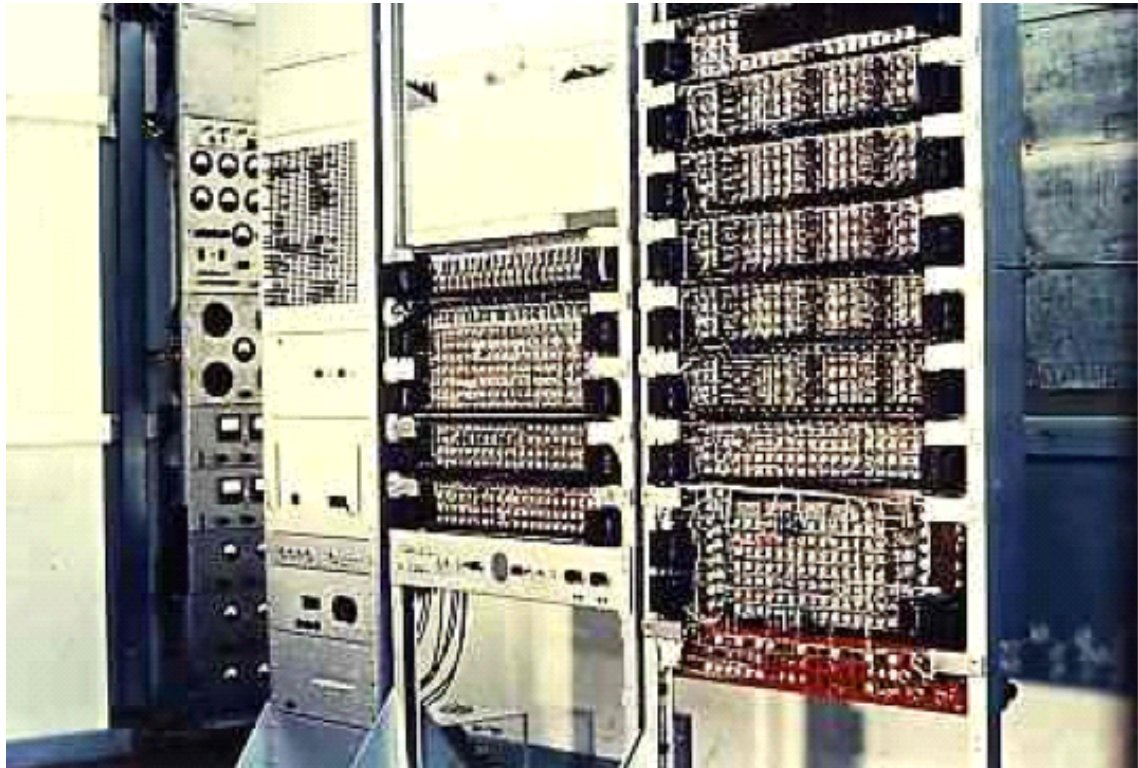
PETER (Philips Elementair Tweetallig Elektronisch Rekenapparaat)

Wim Nijenhuis

1953-1956

Testing

(agreement with
IBM)



History of Science

Dinosaurs: Eindhoven

PASCAL (Philips Akelig Snelle Calculator)

STEVIN (Snel Tel- En Vermenigvuldigings Instrument)



History of Science

Dinosaurs

Note the acronyms and names.

Shell (1953) made the newspapers with their new machine: MIRACLE (Mokums Industrial Research Automatic Calculator for Laboratory and Engineering)

All these people got involved with one another, either in projects (sea defence), in the Colloquium “Moderne Rekenmachines” (since 1952) or in the development of ALGOL 60.

History of Science

Artificial Intelligence

Turing 1950 - Dartmouth 1956

Various strands of research already existing

Research field?

Cybernetics

History of Science

History of AI

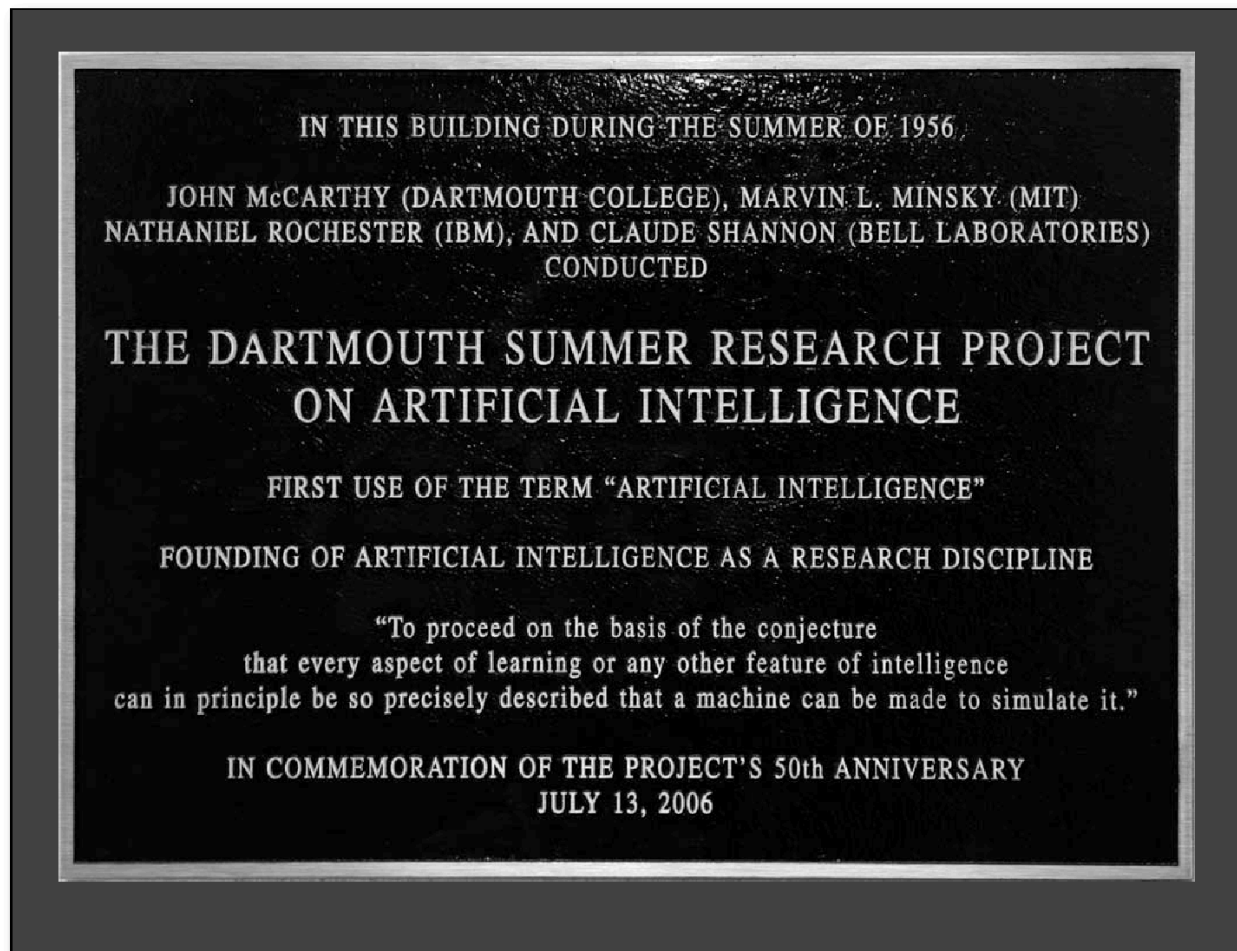
A.M. Turing, *Intelligent Machinery* (1948); Computing machinery and intelligence (1950). Not a research discipline, but...

Cybernetics:

- psychology (cognitive)
- language (learning and translating)
- mathematics - formal logic, proof checking
- philosophy
- biologists (animal behavior, evolutionary and population)
- ...

History of Science

Artificial Intelligence



History of Science

Artificial Intelligence

1956 Dartmouth Conference: The Founding Fathers of AI



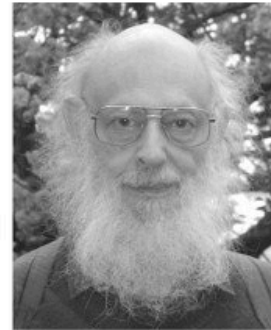
John MacCarthy



Marvin Minsky



Claude Shannon



Ray Solomonoff



Alan Newell



Herbert Simon



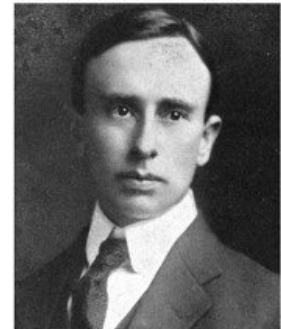
Arthur Samuel



Oliver Selfridge



Nathaniel Rochester



Trenchard More

History of Science

Artificial Intelligence

Caltech / Princeton Mathematics

LISP



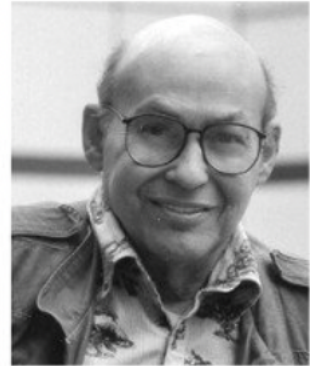
John McCarthy

History of Science

Artificial Intelligence

Harvard / Princeton Mathematics

Society of Mind (with Seymour Papert)



Marvin Minsky

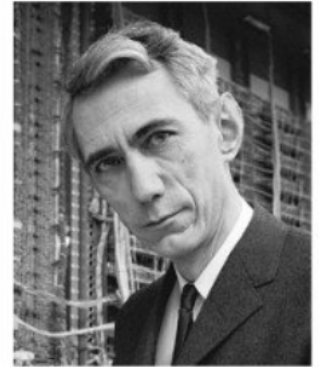
History of Science

Artificial Intelligence

Michigan Mathematics, MIT Electrical engineering

Chess on computer

Mouse Theseus in a maze (automatic switching in telephone circuits)



Claude Shannon



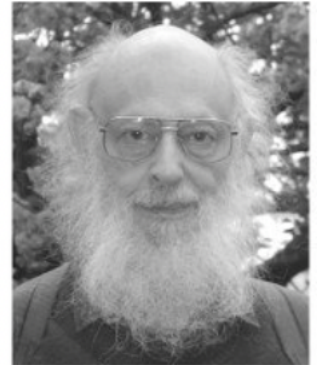
History of Science

Artificial Intelligence

Chicago Physics

Probabilistic grammars

Algorithmic probability (odd one out)



Ray Solomonoff

History of Science

Artificial Intelligence

Chicago, social sciences and
mathematics, political sciences

Econometrics

Administrative behavior (1947)



Herbert Simon

Stanford Physics / Princeton Mathematics

RAND corporation - logistics, game theory



Alan Newell

History of Science

Artificial Intelligence

MIT Electrical engineering

Bell Labs

IBM - checkers

Stanford (1966) TeX project



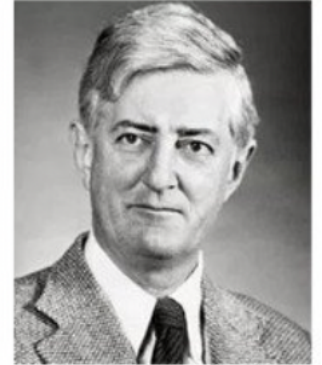
Arthur Samuel

History of Science

Artificial Intelligence

MIT, student of Norbert Wiener

Neural networks



Oliver Selfridge

History of Science

Artificial Intelligence

MIT, Electrical Engineering

IBM 701

1955: pattern recognition study

Developing IBM programming languages



Nathaniel Rochester

History of Science

Artificial Intelligence

Harvard Mathematics - MIT Electrical
engineering



Trenchard More

History of Science

Artificial Intelligence

The “field” of cybernetics was much broader and had very diverging agenda’s

Margaret Mead (1901-1978)

Macy conferences on cybernetics (1946-1953), 1967 keynote ASC

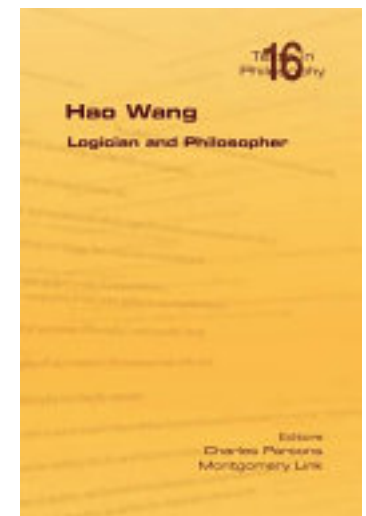
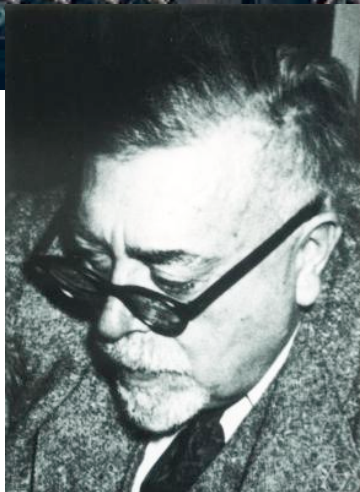


Margaret Masterman (1910-1986)

History of Science

Artificial Intelligence

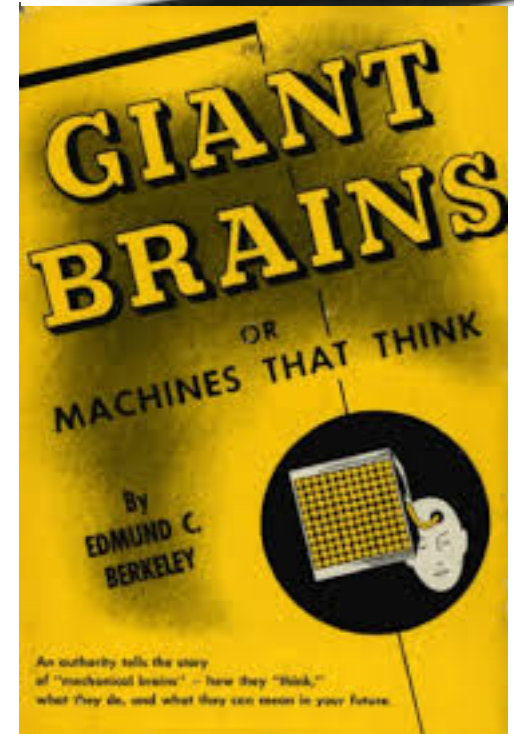
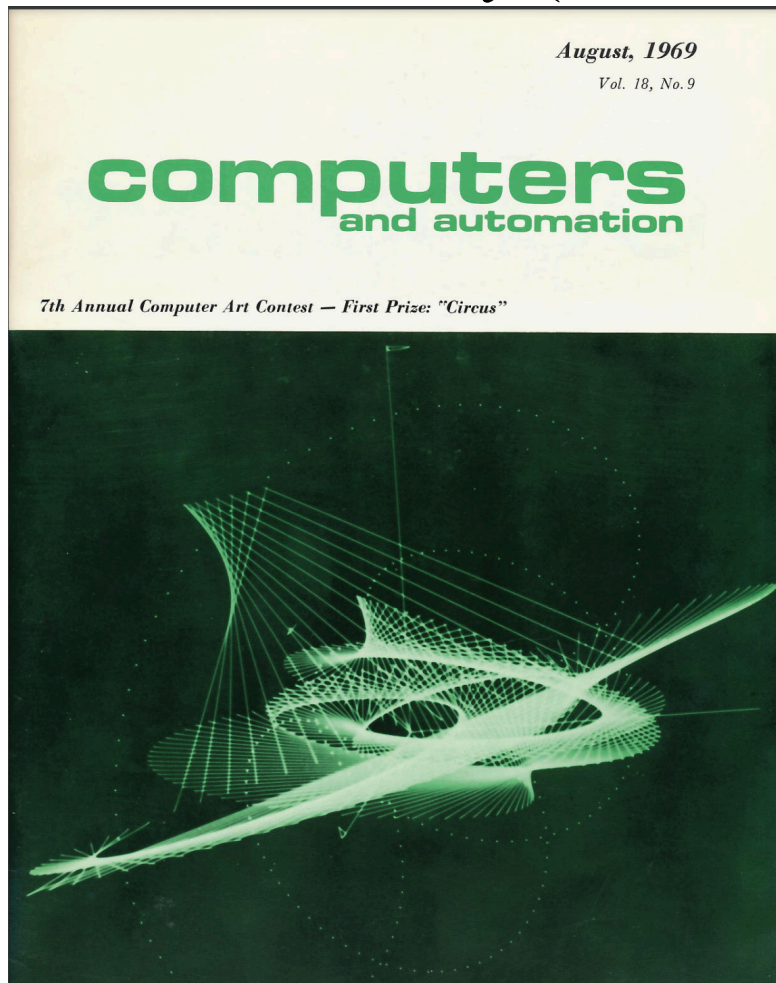
Who were not there?



History of Science

Artificial Intelligence

Edmund Berkely (1909-1988)



History of Science

Artificial Intelligence

January 1951, Paris: conference “Les machines à calculer et la pensée humaine”

From a CS perspective, the conference has retrospectively (2000s) been called “a missed opportunity”



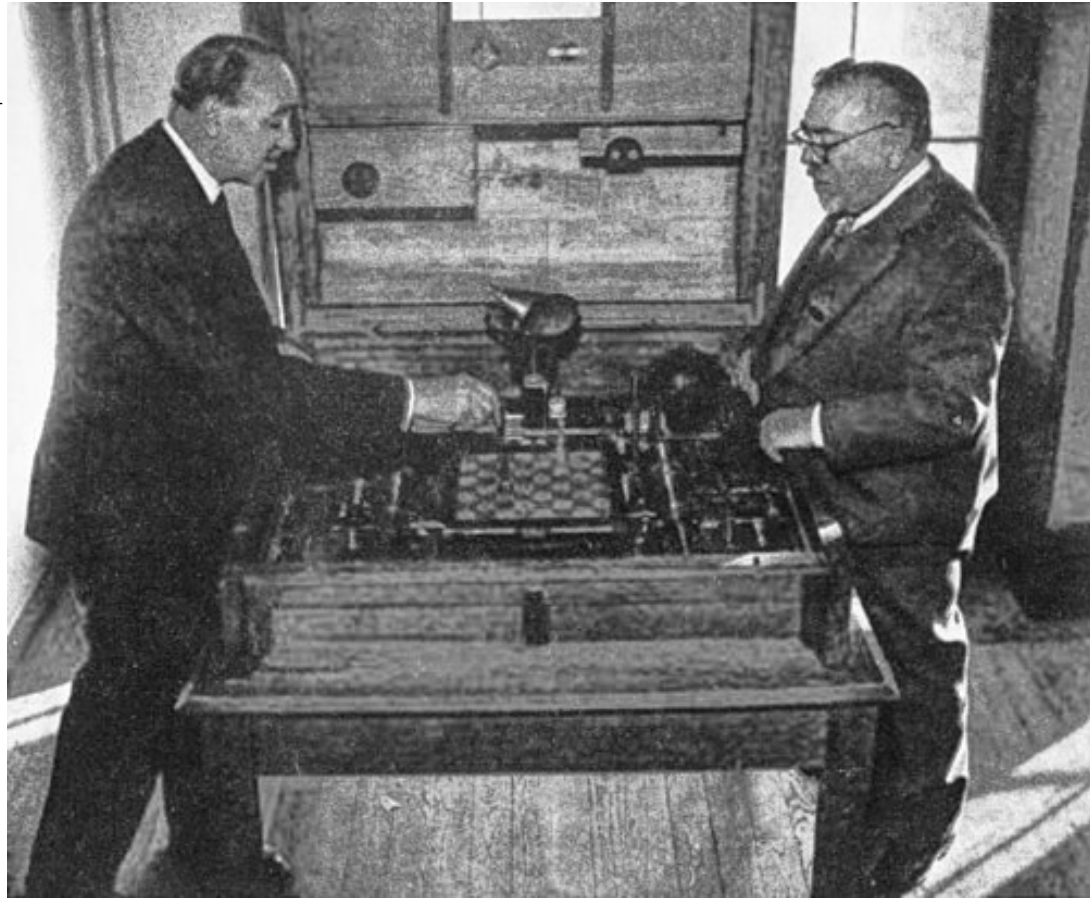
History of Science

Artificial Intelligence

Not present:

Alan Turing

Margareth Masterman



History of Science

Artificial Intelligence

Norbert Wiener

Howard Aiken

Maurice Wilkes

J. Grey Walter / Vivian Dovey (Bristol)

The living brain (1953)

Alwin Walther (Darmstadt)

Adriaan van Wijngaarden

Pierre de Latil (*Thinking by machine*, 1953)

B.V. Bowden (Ferranti, *Faster than thought*, 1953)

lot of female computers among the 270 participants

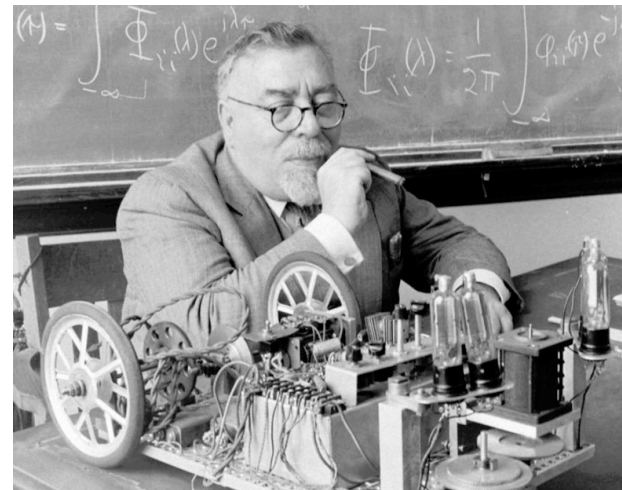
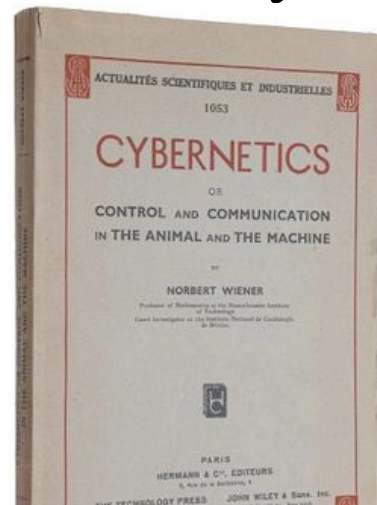


History of Science

Artificial Intelligence

The working of the human brain:

- ▶ in situations of decision making (economics, administration, process control)
- ▶ in automatic translation / learning a language
- ▶ until 1980: in a very deterministic way (Newell / Simon)!



History of Science

Artificial Intelligence

Mathematics and logic (and statistics!)

The hype and the research in 2006
popularly dubbed “the AI winters”

1956 Dartmouth Conference: The Founding Fathers of AI

IN THIS BUILDING DURING THE SUMMER OF 1956
JOHN MCCARTHY (DARTMOUTH COLLEGE), MARVIN L. MINSKY (MIT)
NATHANIEL ROCHESTER (IBM), AND CLAUDE SHANNON (BELL LABORATORIES)
CONDUCTED
THE DARTMOUTH SUMMER RESEARCH PROJECT
ON ARTIFICIAL INTELLIGENCE
FIRST USE OF THE TERM “ARTIFICIAL INTELLIGENCE”
FOUNDING OF ARTIFICIAL INTELLIGENCE AS A RESEARCH DISCIPLINE
“To proceed on the basis of the conjecture
that every aspect of learning or any other feature of intelligence
can in principle be so precisely described that a machine can be made to simulate it.”
IN COMMEMORATION OF THE PROJECT'S 50th ANNIVERSARY
JULY 13, 2006



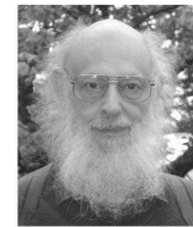
John McCarthy



Marvin Minsky



Claude Shannon



Ray Solomonoff



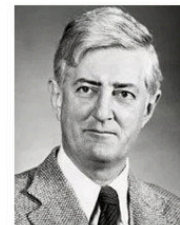
Alan Newell



Herbert Simon



Arthur Samuel



Oliver Selfridge



Nathaniel Rochester



Trenchard More

History of Science

THE END (for now)

Do not forget to:

- Make the assignment
- Hand in the assignment
- Prepare for next lecture

